

False Positives on Free AI Writing Detectors: ELL and Native Essays in School Corpora

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Abstract

The use of artificial intelligence (AI) writing detectors in schools has raised equity concerns that essays by English Language Learners (ELL/ESL) may be disproportionately flagged as AI-generated. Prior work reported high false-positive rates for several GPT detectors when applied to non-native TOEFL essays. The present article tests whether a similar ELL disadvantage appears on free and open detectors applied to school writing corpora rather than to TOEFL-style tasks. Using only human-written essays (any AI or mixed label was coded as a false positive), this study evaluated ZeroGPT's free web interface and the open-source RoBERTa detector across two samples: 200 ASAP 2.0 essays (100 native, 100 ELL) and a high-sensitivity contrast of 100 low-proficiency ELLIPSE essays and 100 length-matched, high-lexical-diversity ASAP native essays. In both samples and for both detectors, ELL essays did not show higher false-positive rates than native essays. Using lexical and syntactic complexity measures, native and ELL essays in this sample did not differ reliably, although RoBERTa false positives were associated with shorter, less lexically diverse text. These results do not support a general ELL false-positive disadvantage for ZeroGPT or this RoBERTa classifier on ASAP 2.0 and ELLIPSE. Fairness risk may depend on the detector and the corpus, not solely on native versus non-native language status. Schools should not treat scores from individual free detectors as evidence of misconduct.

Keywords: AI writing detectors, false-positive rates, English Language Learners, educational equity, detector bias, ASAP 2.0, ELLIPSE, ZeroGPT, RoBERTa

Lede

Free AI writing detectors did not flag English learners' school essays more often than native speakers' essays; errors tracked short, plain text instead, so schools should not trust single detector scores.

Introduction

Generative large language models (LLMs) such as ChatGPT are now widely available to students for brainstorming, drafting support, and revision. In the Higher Education Policy Institute's 2025 UK undergraduate survey, 92% of students reported using AI in some form and 88% reported using generative artificial intelligence (AI) for assessments, up from 66% and 53% a year earlier (Freeman, 2025). Many schools and universities have responded by adopting commercial or free AI writing detectors, including GPTZero, Turnitin, Copyleaks, ZeroGPT, and open-source classifiers, to screen student work (Weber-Wulff et al., 2023; Center for Democracy & Technology [CDT], 2024). These tools typically score "AI-likeness" from statistical or neural signals. Early versions of GPTZero, for example, publicized perplexity and burstiness as core cues (Tian, 2023). Because current language models are trained to resemble human prose, the margins on which such signals rely are often narrow, and ordinary human writing can cross a flagging threshold (Heikkilä, 2023; Weber-Wulff et al., 2023). Fowler (2023) tested Turnitin's detector on real high-school writing; the system flagged part of a fully human-authored student essay as AI-generated. A false positive can trigger an integrity conversation or formal investigation before a teacher actually reads the essay on its merits.

If a detector systematically misclassifies some student groups more than others, it can produce unequal academic consequences even without intentional discrimination. English language learners (ELLs) are the group most often discussed. Learner writing often differs from native-speaker writing in lexical range, syntactic complexity, and error profiles—features that some detectors treat as markers of "AI-like" predictability (Liang et al., 2023). Civil-society analyses have warned that English learners may face unequal discipline if schools treat detector scores as proof of AI use or plagiarism (CDT, 2023, 2024). Schools therefore need evidence on which detectors, and which textual features, actually raise ELL false positives, since researchers cannot assume that every detector shares the same bias profile.

Liang et al. (2023) provided the most influential evidence for this concern, reporting that several GPT detectors flagged non-native TOEFL essays as AI-generated far more often than essays by U.S. eighth-grade students—a

finding that many schools now treat as urgent. Whether the same pattern holds outside the detectors and tasks that study examined is the gap this paper addresses.

This research addresses three questions. RQ1. On same-task ASAP 2.0 school essays—a U.S. secondary source-based argumentative corpus documented by Crossley et al. (2025) and publicly distributed via Kaggle—do ZeroGPT and an open RoBERTa AI-writing detector produce higher false-positive rates for ELL essays than for native essays? RQ2. In a pre-specified high-sensitivity contrast (Sampling v2: low Overall/low MTLD/low Vocabulary ELLIPSE essays versus length-matched high-TTR ASAP natives), is an ELL false-positive disadvantage more detectable? RQ3. Which linguistic features are associated with false positives on these free and open detectors?

This study examines everyday school writing and the free and open-source tools that teachers can use directly, to test whether the same risk holds there. The results can give schools more specific evidence about whether and how much to rely on such detectors, rather than applying the conclusions of one prior study to every tool and every writing task.

Literature Review

Detector Architectures

AI writing detectors are often treated as a unified category, but the underlying mechanisms differ. Commercial web tools such as GPTZero and ZeroGPT mainly score statistical features of the text, most commonly perplexity (how predictable a language model's word choices are in the text) and burstiness (sentence-level variation). Texts with smooth language, limited structural variation, and a relatively uniform rhythm may receive higher AI scores even when those features do not come from AI generation (Tian, 2023). In contrast, open-source encoder models such as Hello-SimpleAI's chatgpt-detector-roberta fine-tune RoBERTa on labeled human and AI text, so the model can learn more complex patterns from vocabulary, sentences, and context rather than relying on a few pre-defined statistical indicators (Guo et al., 2023). Because the two types of detectors rely on different signals, they may also exhibit different error patterns. This study therefore examined a commercial-style web detector (ZeroGPT) and an open-source encoder classifier (RoBERTa) rather than assuming that bias observed in one detector generalizes to another.

Bias Against Non-Native Writing

The concern that AI writing detectors penalize non-native English writing began with Liang et al. (2023), who evaluated seven GPT detectors on 91 human-authored TOEFL essays and 88 U.S. eighth-grade essays. The average false-positive rate was about 61% on the TOEFL set, against near-perfect accuracy on the U.S. student essays; all seven detectors agreed on 19.8% of the TOEFL essays, and at least one detector flagged 97.8% of that set. The Center for Democracy & Technology (2023, 2024) drew on this evidence to warn that English learners may face unequal discipline if schools treat detector scores as proof of AI use or plagiarism.

Subsequent work has qualified rather than confirmed that pattern. Jiang et al. (2024) examined first-language (L1) and second-language (L2) writing drawn from the same large-scale testing program and did not consistently find an ELL penalty. Because their comparison held task and testing context constant in a way the TOEFL-versus-eighth-grade contrast did not, it separates nativeness from task and proficiency more cleanly. Read together, the two studies describe a trajectory: a highly influential alarm using TOEFL-style independent writing, followed by a qualification using authentic school-assessment writing. Before an “ELL penalty” is treated as a general property of AI detectors, the claim needs to be tested on the free and open tools schools actually use, applied to everyday school writing.

Linguistic Features

Liang et al. (2023) found that false-positive rates were associated with lower perplexity and lower lexical diversity: when TOEFL essays were revised to be closer to native-language writing, the false-positive rate decreased; when native speakers' compositions were simplified, the false-positive rate rose. In second language acquisition research, however, a limited vocabulary range, shorter or simpler syntactic structures, and non-standard usage are well-documented L2 writing features and are not unique to AI-generated text. Learner corpus research treats these patterns as systematic features of interlanguage rather than as signals of non-human authorship (Granger, 1998).

If the cues relied on by a detector overlap with the language features used in SLA studies to describe second-language development, the detector may have difficulty distinguishing learner writing from AI-generated text. This overlap is not unique to ELL writers: native English speakers may also use shorter sentences, repetitive words, or frequent clichés. Misreading normal second-language development as AI writing is therefore both a measurement problem and an equity risk. This study examines lexical diversity (TTR and MTLD), sentence-level complexity, and a spelling-error proxy, and compares their performance across language groups and false-positive status.

Reliability and Equity

Public detectors are also unreliable in general, independent of any group-based bias. Weber-Wulff et al. (2023) tested 14 tools and found that none reached 80% overall accuracy, with errors increasing after paraphrasing or machine translation. OpenAI has stated that current detectors do not reliably separate AI-generated from human-generated text (OpenAI, 2023). Because false accusations can affect grades and disciplinary records, this baseline unreliability is itself a reason to require human review and to test tools on the students a school actually serves.

Two questions remain open. First, it is not yet known whether a commercial-style web detector and an open-source encoder classifier share the same ELL bias profile when applied to the same essays. Second, the trajectory from Liang et al. to Jiang et al. has not been extended to ZeroGPT and an open RoBERTa classifier on authentic school writing with a purpose-built contrast designed to make any ELL disadvantage easier to detect. This study addresses both gaps. It does not seek to contradict Liang et al. (2023), but tests whether their finding extends to free and open detectors when the object of analysis is authentic school essays.

Materials and Methods

Research Design

This study is a comparative secondary analysis of authentic human-authored essays, with a secondary analysis of linguistic features. The primary independent variable is writer group (native vs. ELL). The dependent variables are the detector's label or AI probability and the group false-positive rate (FPR). All sampled texts are human-written, so any non-human label is a false positive. No essays were rewritten, AI-edited, or generated for this study. Sampling rules for the high-sensitivity follow-up were fixed before those essays were scored.

Corpora

The primary corpus is ASAP 2.0, a U.S. secondary source-based argumentative assessment that records ELL status and prompt IDs, so native and ELL essays can be compared within the same task (Crossley et al., 2025). ELLIPSE provides independent prompts and proficiency trait scores and is used only to build the low-proficiency ELL subsample in Sampling v2 (Crossley et al., 2023). LOCNESS provides extra native writing for exploratory checks across genre and age; LOCNESS–ELLIPSE contrasts are not treated as native–ELL tests. Only the student essay body was sent to detectors.

Sampling

Sampling v1 — ASAP 2.0 (n = 200)

One hundred native and 100 ELL essays were drawn from ASAP 2.0, each 150–400 words. Essays were balanced across the seven ASAP prompts (about 14–15 per prompt per group). Mean length was 260.9 words for native essays (range 160–393) and 270.9 for ELL essays (range 152–396).

Sampling v2 — High-Sensitivity Contrast (n = 200)

The ASAP v1 ZeroGPT contrast was null (FPR = 1% in both groups). Sampling v2 therefore targeted the kind of text prior work links to higher false positives—lower proficiency and lower lexical diversity—without creating new text. The ELL set consists of 100 ELLIPSE essays of 150–400 words with Overall ≤ 3.0 , MTLD ≤ 40 , and Vocabulary ≤ 3.0 (mean Overall = 2.52; mean MTLD = 34.20; mean length = 291.04). The native set consists of 100 ASAP 2.0 non-ELL essays in the same length band, with simple TTR at or above the ASAP native median, length-bin matched to the ELL set, and no overlap with Sampling v1 (mean TTR = 0.539; mean length = 291.37).

LOCNESS native samples (150–400 words, $n = 71$; 150–600 words, $n = 100$) were scored only as an exploratory check.

Detectors and Coding

Two detectors were scored on both samples. A false positive is any human essay labeled non-human under the rule below. Essays that could not be parsed were dropped from that detector's FPR denominator.

ZeroGPT (Free Web UI)

The headline label and AI percentage were recorded in August 2026. Labels of AI or mixed were coded as false positives; human labels were coded as 0. Unclear outputs were excluded. Free web tools change without notice, so these scores are tied to that collection window.

Open-Source RoBERTa Detector

The model is Hello-SimpleAI/chatgpt-detector-roberta (Guo et al., 2023), run locally. An essay was labeled AI if $P(\text{AI}) \geq .50$. This model has no mixed class.

GPTZero

A 50-essay ASAP v1 pack was prepared but not fully scored. GPTZero is therefore omitted from confirmatory tables.

Procedure

Essays were sampled under the rules above, given study IDs, and exported as one UTF-8 text file each, containing only student prose. Packaging noise was removed; learner errors were not corrected. The same files were submitted to ZeroGPT and to the local RoBERTa model. For each essay and detector, the dataset records group, corpus, label, AI probability or AI percentage, false-positive code, word count, and detection date.

Linguistic Features

Features are used for sampling checks and for explanatory contrasts (native vs. ELL; false positive vs. not). They are not the primary hypothesis test and are not treated as causes of detector error. The measures are word count, TTR, MTLD, mean sentence length, and a spelling-error proxy.

Word count (N) is the number of whitespace-separated tokens in the essay body. Simple type-token ratio (TTR) is V/N , where V is the number of unique word types. TTR falls as texts get longer, so it is reported together with length and is used in Sampling v2 only after essays are already restricted to 150–400 words. MTLD (Measure of Textual Lexical Diversity; McCarthy & Jarvis, 2010) is the mean length of successive word strings that maintain a TTR of 0.72. Each essay was scanned forward and then backward; the two factor lengths were averaged. Unlike raw TTR, MTLD is less sensitive to text length, which is why Sampling v2 used $\text{MTLD} \leq 40$ to select lexically restricted ELL essays. Mean sentence length is N/S , where S is the number of sentences split by end-of-sentence punctuation. The spelling-error proxy is the rate of dictionary-unknown tokens ($U/N \times 100$), using the pyspellchecker dictionary. This is an unknown-word rate, not a full grammar-error score; proper nouns and prompt-specific names can also be marked unknown.

Analytical Strategy

For each detector and sample, the study reports group FPR; mean AI percentage (ZeroGPT) or mean $P(\text{AI})$ (RoBERTa); a 2×2 test of group $\times$ false-positive status; and an odds ratio for ELL versus native, with a 95% confidence interval. The primary test is Pearson's chi-square. When any expected cell is below 5, Fisher's exact test is used instead. An odds ratio greater than 1 means higher odds of a false positive for ELL essays. Feature means are compared by group. All tests are two-sided with $\alpha = .05$.

Ethics and Limitations

The analyses use public educational corpora and report aggregate statistics only. Detector outputs are used only as research measures, not as accusations against writers. Confirmatory results cover only ZeroGPT's free interface and one RoBERTa classifier. Web detectors change outside the researcher's control. Sampling v2 is cross-corpus. The spelling feature is an unknown-word proxy, not a full error tagger. Secondary corpora confound task, variety, and educational level. The study describes disparate impact, not intent.

Results

Four comparisons were completed: ZeroGPT and RoBERTa on ASAP v1, and the same two detectors on the v2 high-sensitivity sample (Table 1). A false positive is an AI or mixed label for ZeroGPT, or $P(\text{AI}) \geq .50$ for RoBERTa. Tests are two-sided. The question for RQ1 and RQ2 is whether ELL FPR exceeds native FPR.

Table 1

False Positives by Sample, Detector, and Group

Sample	Detector	Group	N	FP	FPR	Mean AI score
ASAP v1	ZeroGPT	Native	100	1	1.0%	2.01%
ASAP v1	ZeroGPT	ELL	100	1	1.0%	0.77%
ASAP v1	RoBERTa	Native	100	26	26.0%	29.05%
ASAP v1	RoBERTa	ELL	100	20	20.0%	22.60%
v2 contrast	ZeroGPT	Native	100	5	5.0%	3.01%
v2 contrast	ZeroGPT	ELL	100	0	0.0%	0.74%
v2 contrast	RoBERTa	Native	100	14	14.0%	17.04%
v2 contrast	RoBERTa	ELL	100	9	9.0%	10.26%

Note. ELL – native FPR differences: ASAP v1 ZeroGPT = 0.0 percentage points, OR = 1.00, 95% CI [0.062, 16.213], $\chi^2 p = 1.00$, Fisher $p = 1.00$; ASAP v1 RoBERTa = –6.0, OR = 0.712, 95% CI [0.367, 1.381], $\chi^2 = 0.706$, $p = .401$, Fisher $p = .401$; v2 ZeroGPT = –5.0, OR = 0.086, 95% CI [0.005, 1.584], $\chi^2 p \approx .070$, Fisher $p \approx .059$; v2 RoBERTa = –5.0, OR = 0.608, 95% CI [0.250, 1.476], $\chi^2 p = .375$, Fisher $p = .376$. ZeroGPT label detail: ASAP v1 native, 99 human and 1 AI; ASAP v1 ELL, 99 human and 1 mixed; v2 native, 95 human and 5 mixed; v2 ELL, 100 human. Mean AI score is mean AI percentage for ZeroGPT and mean $P(\text{AI})$ for RoBERTa. Created by author, 2026.

ZeroGPT almost never flagged either group on ASAP v1. The FPR difference was 0.0 percentage points, but that zero should not be read as a precise estimate of no effect. With only one false positive in each group of 100, statistical power is low, and the 95% confidence interval for the odds ratio [0.062, 16.213] is wide enough to accommodate either a substantial ELL disadvantage or a native disadvantage. The contrast is too imprecise to rule out a small effect.

RoBERTa base rates on the same essays are far higher. The native rate is numerically higher, not lower, and the contrast is not significant.

The Sampling v2 contrast was built to make an ELL disadvantage easier to see. ZeroGPT still did not flag ELL essays more often; the trend runs in the opposite direction and does not reach $p < .05$. RoBERTa's overall rates fall relative to ASAP v1, but the native–ELL pattern is unchanged: no ELL elevation. Across both samples and both detectors, ELL FPR was never higher than native FPR. Detector choice changes the absolute rate (ZeroGPT 0–5%; RoBERTa 9–26%); group membership does not.

Exploratory Secondary Corpora (RoBERTa)

These are not confirmatory native–ELL tests. RoBERTa FPR was 7.5% on a 40-essay ELLIPSE subset, 2.8% on LOCNESS essays of 150–400 words ($n = 71$), and 1.0% on LOCNESS essays of 150–600 words ($n = 100$). The LOCNESS native rate (1%) was far lower than the ASAP v1 native rate (26%). Detector behavior therefore changes with corpus and task even among native writers.

Linguistic Features (RQ3; ASAP v1)

Table 2

ASAP v1 Feature Means by Group

Feature	Native M (SD)	ELL M (SD)	ELL – Nat	p
Word count	260.2 (62.3)	269.8 (68.4)	+9.6	.32
TTR	0.515 (0.070)	0.511 (0.078)	–0.005	.56

Feature	Native M (SD)	ELL M (SD)	ELL – Nat	p
MTLD	71.9 (22.6)	72.2 (23.6)	+0.4	.90
Mean sentence length	19.5 (6.1)	22.3 (11.9)	+2.8	.22
Spelling-error density (/100)	2.73 (1.90)	2.93 (2.05)	+0.19	.39

Note. Native and ELL essays on ASAP v1 did not differ reliably on these measures. Created by author, 2026.

Table 3

ASAP v1 Features by RoBERTa False-Positive Status (Groups Pooled)

Feature	FP M (SD)	Non-FP M (SD)	Non-FP – FP	p
Word count	243.9 (47.5)	271.3 (68.9)	+27.4	.023
TTR	0.485 (0.060)	0.521 (0.076)	+0.037	.002
MTLD	63.9 (16.5)	74.5 (24.2)	+10.6	.006
Mean sentence length	20.1 (5.5)	21.1 (10.4)	+1.0	.53
Spelling-error density (/100)	1.65 (1.03)	3.18 (2.06)	+1.53	<.001

Note. RoBERTa false positives were shorter, less lexically diverse, and had fewer dictionary-unknown tokens. Created by author, 2026.

These associations do not show that low diversity causes the error, and length may confound the lexical measures. They do show that, on this corpus, misclassification tracks text properties more closely than the ELL label. That reading is consistent with Sampling v2: low-MTLD, low-proficiency ELL essays still did not produce a higher FPR.

Discussion

On ZeroGPT and the open RoBERTa classifier, ELL school essays were not flagged as AI-generated more often than native essays. That result does not overturn Liang et al. (2023). It shows that the ELL penalty they reported for several GPT detectors on TOEFL writing does not automatically appear for these two free or open tools on ASAP 2.0 and ELLIPSE.

One reason the findings differ is timing. Liang et al. used detectors available in early 2023; the ZeroGPT scores in this study were collected in August 2026. After that paper circulated widely, several commercial companies said they would try to reduce false positives on non-native writing. Free web tools and open-source models need not have received the same updates, but they also need not have stayed unchanged for three years. A 2026 result with no ELL elevation is therefore compatible with at least two explanations. First, the tools themselves may have changed: thresholds or training data may no longer flag learner writing as readily as they did in 2023. Second, the objects of comparison are not the same. The open-source RoBERTa model is an encoder classifier, not one of the commercial GPT detectors Liang et al. tested, and ASAP and ELLIPSE school essays are not TOEFL independent writing. This design can only show that an ELL disadvantage is not a fixed property that every later detector inherits.

The present comparisons are also narrower. Sampling v1 contrasts native and ELL essays from the same ASAP assessment; Sampling v2 is a harder ELLIPSE-versus-ASAP contrast. Across those four tests, the ELL–native difference was zero or negative. Two patterns matter more than any single p value. First, the overall false-positive rate depends on the tool: ZeroGPT stayed between 0% and 5%, while RoBERTa stayed between 9% and 26%. Second, Sampling v2 was built to make an ELL disadvantage easier to detect, and it still did not appear. That contrast, however, is cross-corpus. Prompt wording, source use, and writing conditions differ between ELLIPSE and ASAP even after length and lexical diversity are aligned, so v2 is a sensitivity check rather than a pure native–ELL experiment. The cleaner group comparison is the same-task ASAP sample. Even there, ZeroGPT produced so few false positives that the group contrast remains imprecise.

The feature analysis helps explain why some essays were still mislabeled even though the two groups had similar false-positive rates. On the same-task ASAP sample, native and ELL essays did not differ on length, TTR, MTLD, sentence length, or the spelling proxy. The comparison was therefore not already stacked with simpler ELL text. Regardless of L1 or L2 status, shorter essays with lower lexical diversity were more likely to be flagged by RoBERTa. Errors tracked these stylistic markers, not whether the writer was ELL. False-positive essays also

contained fewer dictionary-unknown tokens. That pattern may reflect plainer, more common vocabulary rather than better spelling. The result is partly consistent with Liang et al.'s predictability account, because low lexical diversity was associated with error. It is inconsistent in another respect: those features were not concentrated among ELL writers, so they did not produce an ELL gap.

The confirmatory analysis used only two detectors, and ZeroGPT's website algorithm may change without notice, so the current scores may not be fully reproducible. None of the four statistical tests underwent multiple-comparison correction. The results should therefore be interpreted with caution. A clear next step is to re-score the same papers using the detection systems that schools actually employ for assignments and academic-integrity review, including GPTZero, Turnitin, and newer classroom tools such as Pangram. Re-testing the same ASAP essays can show whether the lack of a group difference observed here is limited to free or open-source detectors or also applies to commercial systems used in integrity decisions.

Conclusion

This replication–extension study finds that ZeroGPT and an open RoBERTa classifier did not flag ELL school essays as AI-generated more often than native essays, either on ASAP 2.0 or on a high-sensitivity ELLIPSE/ASAP contrast. An ELL false-positive disadvantage is not something every detector automatically produces. RoBERTa errors were associated with shorter, less diverse, and orthographically cleaner text rather than with ELL status. The association does not directly show that those features cause the error, but it does suggest that risk tracks textual properties more closely than group labels. The risk depends on which tool is used and on which writing is scored. Schools should check a detector on the assignments they actually give and keep a teacher's reading at the center of any integrity decision. They should not treat a single detector score as proof of misconduct.

Appendix

The public archive at <https://github.com/isabellayu1208-dotcom/ai-detector-ell-bias> contains the materials needed to check the tables in this paper without redistributing student essays. The data folder includes sampling metadata and study IDs (essay bodies are not included). The docs folder contains the pre-specified rules for the high-sensitivity ELLIPSE/ASAP contrast. The results folder contains detector labels and confirmatory summaries. The scripts folder contains Python used to compute word count, TTR, MTLT, mean sentence length, and the pyspellchecker unknown-token rate. Readers who obtain ASAP 2.0 and ELLIPSE from the official providers can recompute the feature tables locally.

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